Enzymes

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Intended Learning Outcomes

1. Explain how enzymes function as highly selective biological catalysts
2. Describe the three-dimensional structural anatomy of an enzyme's active site and explain how its microenvironment provides substrate specificity.
3. Characterize the operational properties of regulatory enzymes
4. Discuss the significance of non-protein biological catalysts

What are Enzymes?

- Enzymes are biological catalysts.

- This means they are substances that increase the rate of a biochemical reaction without being consumed or permanently altered in the process.

- They are central to virtually every metabolic process, from digestion and energy production to DNA replication and immune responses.

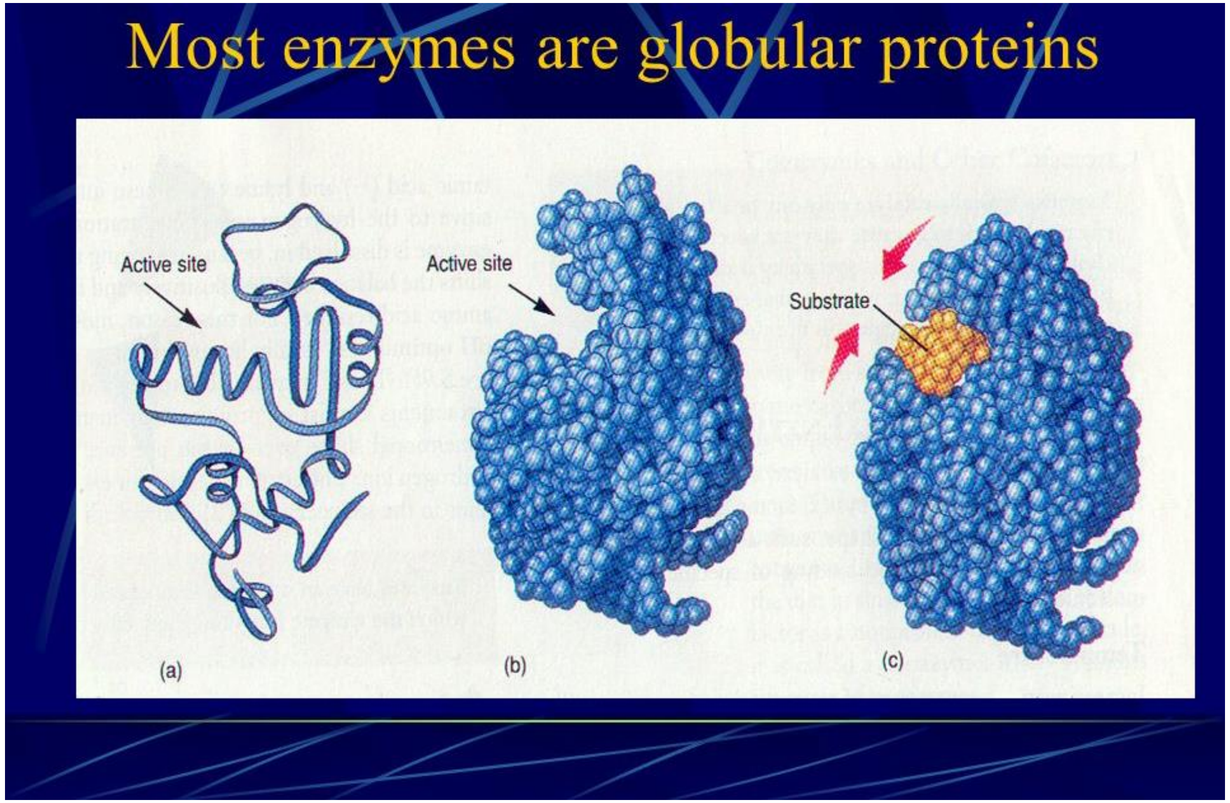
Key Characteristics

- **Catalytic Power:** They can accelerate reaction rates by enormous factors (up to 1017 times faster than uncatalyzed reactions!)
- **Specificity:** They are highly selective, acting on very specific substrates to produce specific products
- **Regulation:** Their activity can be precisely controlled, allowing cells to respond to changing needs and maintain homeostasis

The Nature of Enzymes

Globular Proteins:

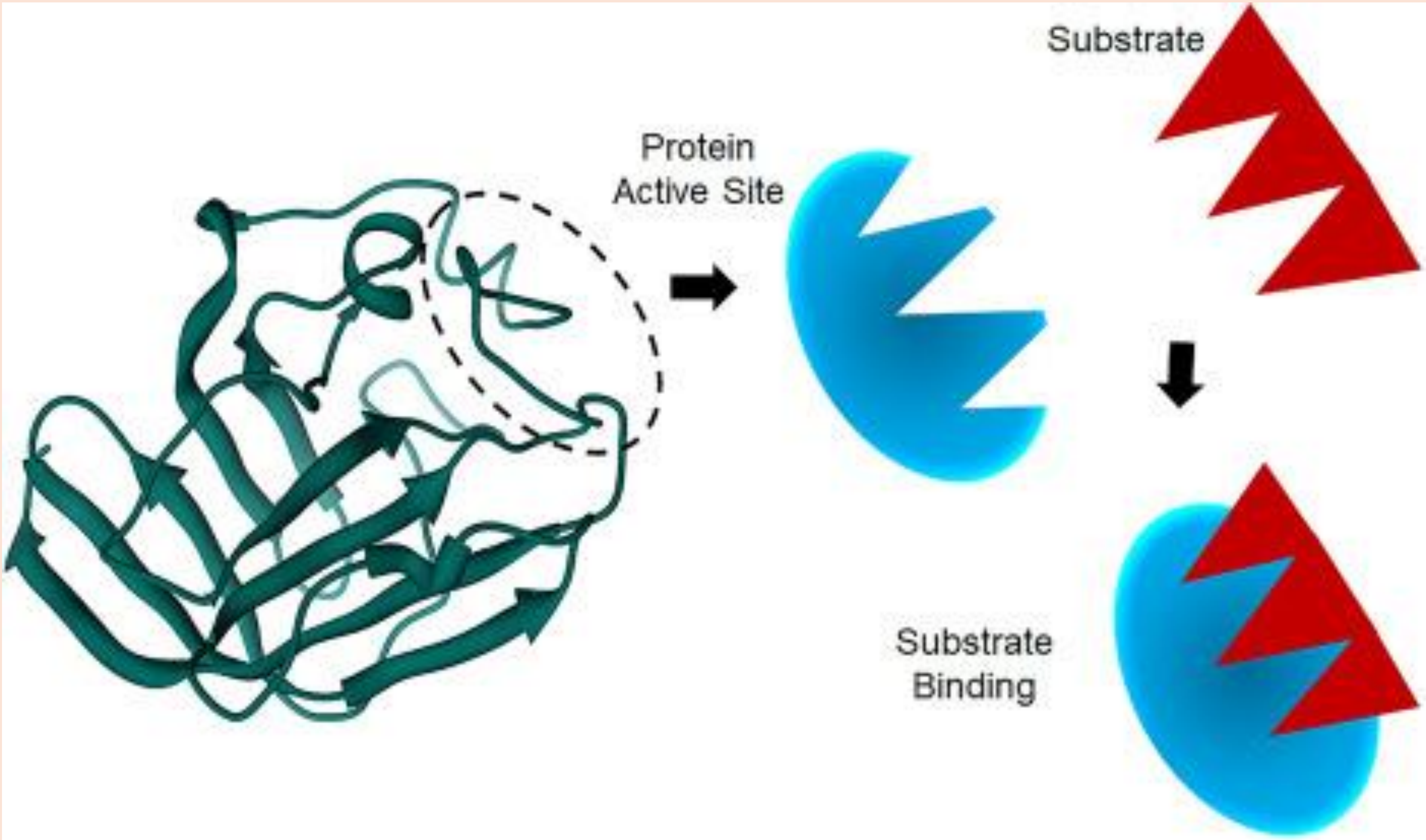
- Most enzymes are globular proteins, meaning they have a compact, roughly spherical three-dimensional structure.



The Nature of Enzymes

Active Site:

- The most crucial part of an enzyme is its **active site.**
- This is a specific, three-dimensional pocket or cleft on the enzyme's surface (or within its structure) where the substrate binds and the chemical reaction occurs.
- It's formed by the specific arrangement of a few amino acid residues, which may be far apart in the primary sequence but are brought close together by protein folding.
- The active site's shape and chemical environment are highly specific for the enzyme's particular substrate(s).



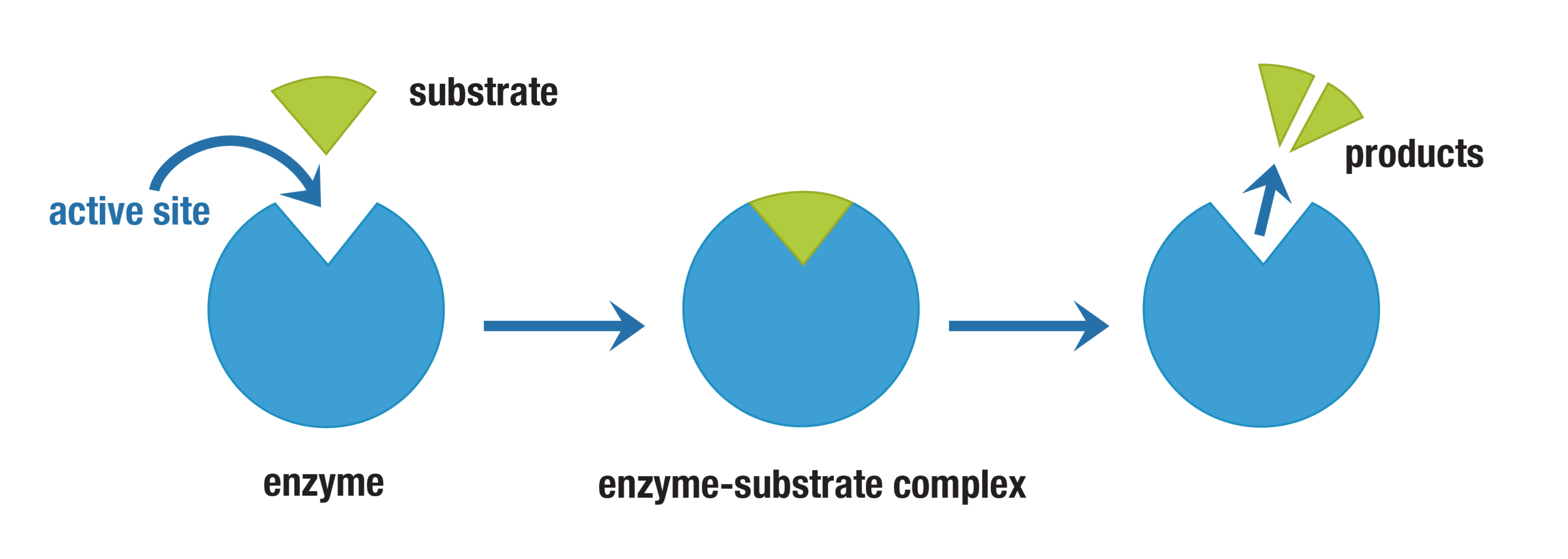
Protein
Active Site
Substrate
Substrate
Binding

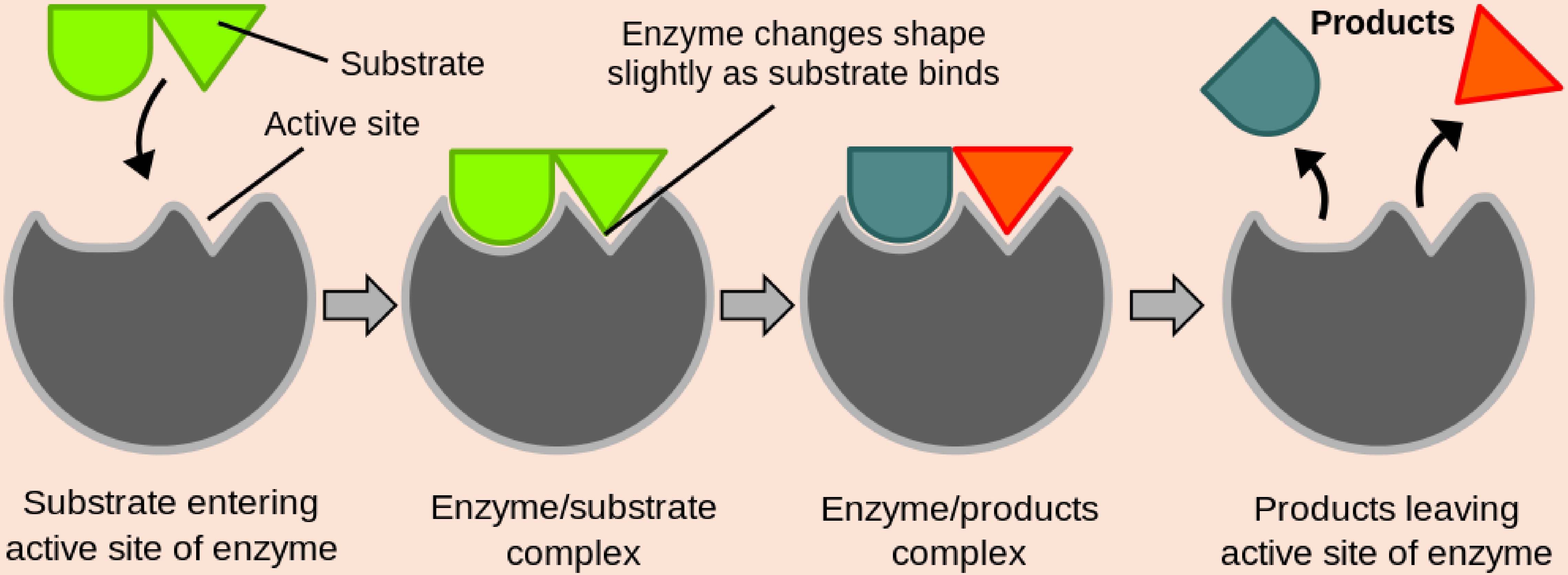
The Nature of Enzymes

Enzyme-Substrate (ES) Complex:

- When the substrate binds to the active site, an intermediate **enzyme-substrate complex (ES complex)** is formed.

- This binding is typically non-covalent (hydrogen bonds, ionic interactions, hydrophobic interactions, Van der Waals forces).

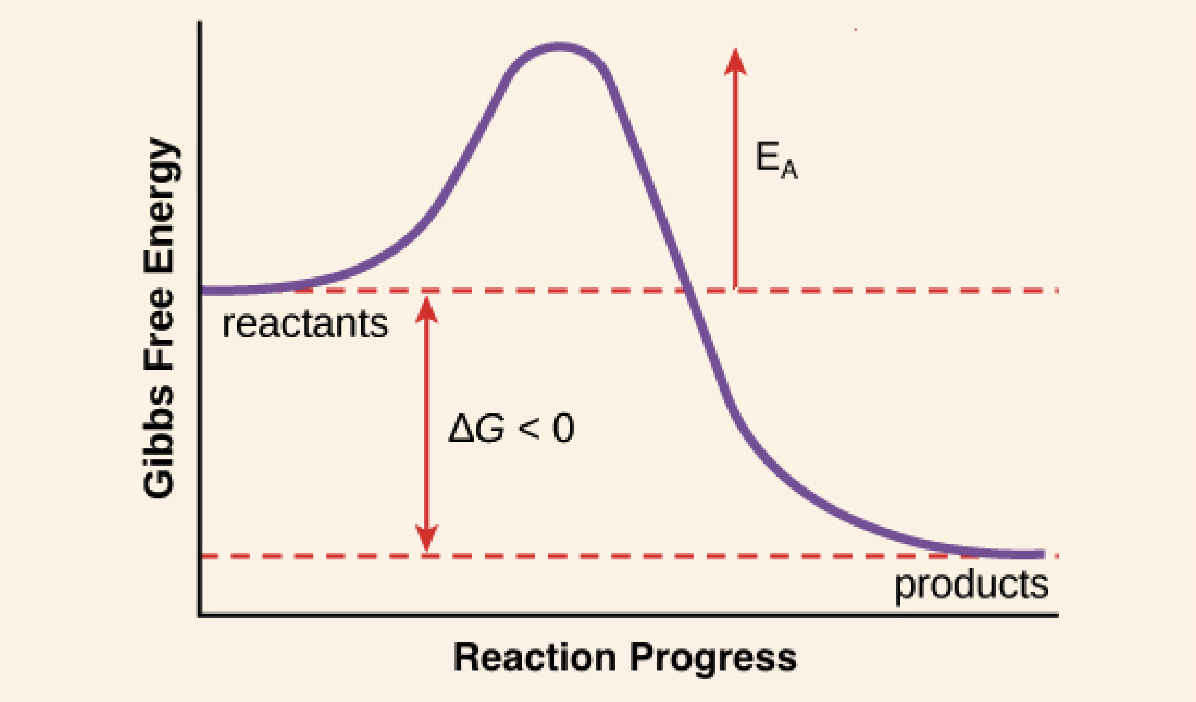

substrate
active site
products
enzyme
enzyme-substrate complex


Substrate
Active site
Enzyme changes shape slightly as substrate binds
Products
Substrate entering active site of enzyme
Enzyme/substrate complex
Enzyme/products complex
Products leaving active site of enzyme

How Enzymes Work

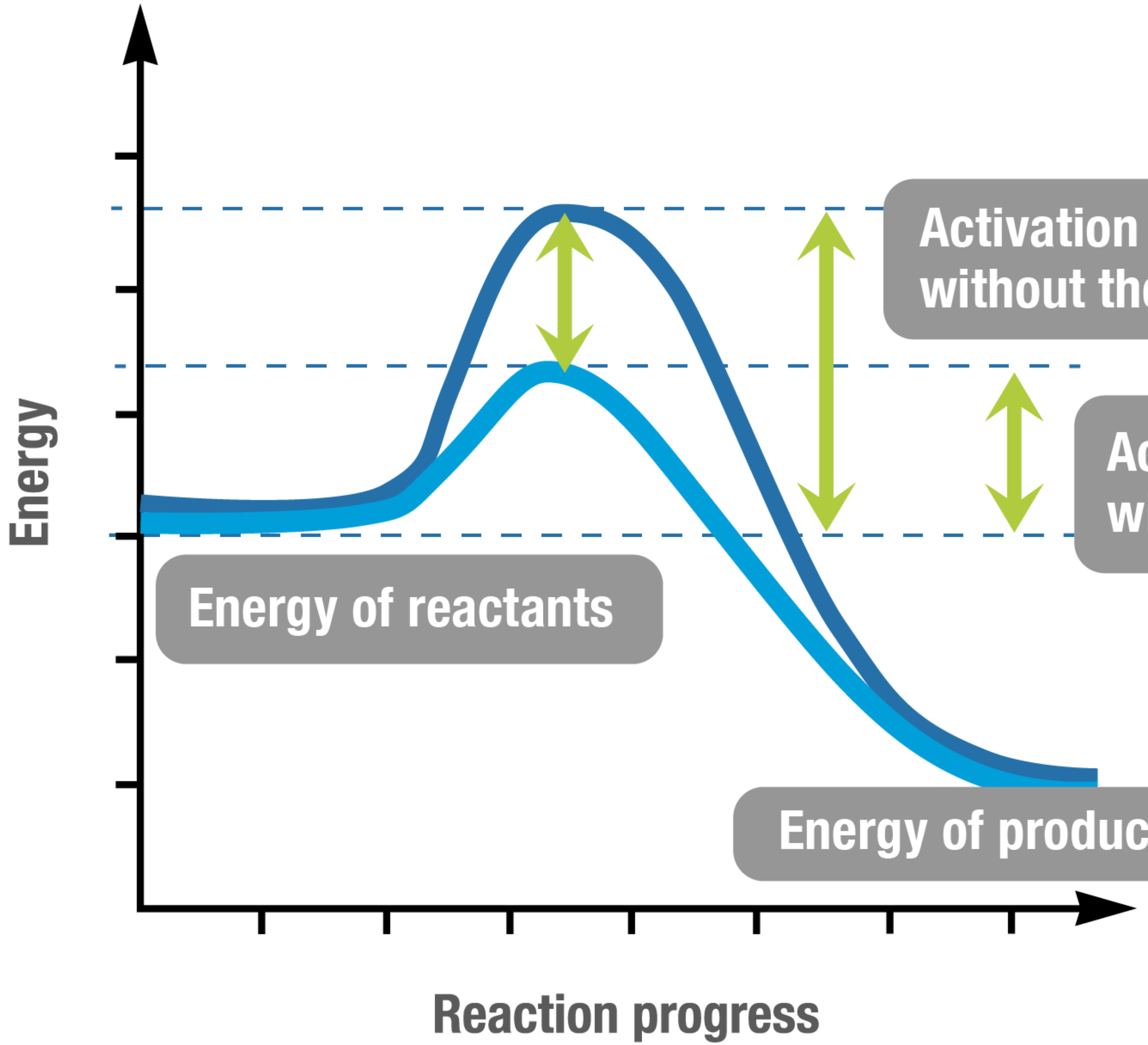
- Chemical reactions require a certain amount of energy input to get started, known as the **activation energy (Ea)**.

- This energy is needed to reach a high-energy, unstable **transition state** where bonds are breaking and forming.



Enzymes accelerate reactions by lowering the activation energy (Ea) of the reaction. They do this by:

- **Stabilizing the transition state:** The active site binds to the transition state much more tightly than to the substrate or product, effectively lowering its energy.
- **Providing an optimal environment:** Creating a microenvironment within the active site that facilitates the reaction (e.g., specific pH, exclusion of water).
- **Orienting substrates:** Bringing substrates into precise proximity and orientation for the reaction to occur.
- **Straining substrates:** Inducing conformational changes in the substrate that push it towards the transition state.


Energy
Activation energy without the Enzyme
Activation energy with the Enzyme
Energy of reactants
Energy of products
Reaction progress

How Enzymes Work

Key Principle

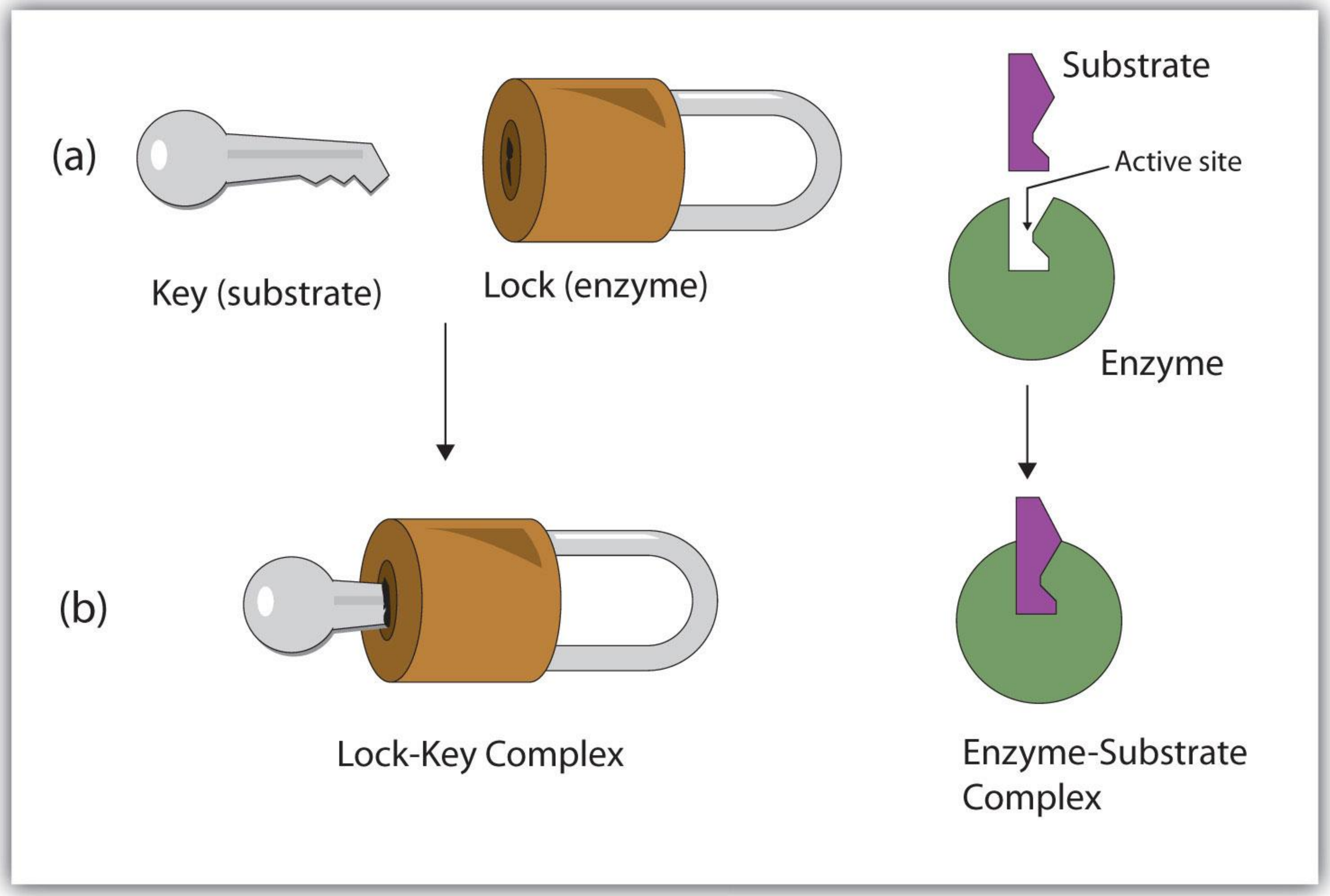
- Enzymes **do not alter the equilibrium (Keq)** of a reaction.
- They only speed up the rate at which equilibrium is reached.
- The same amount of product will eventually form, just much faster.

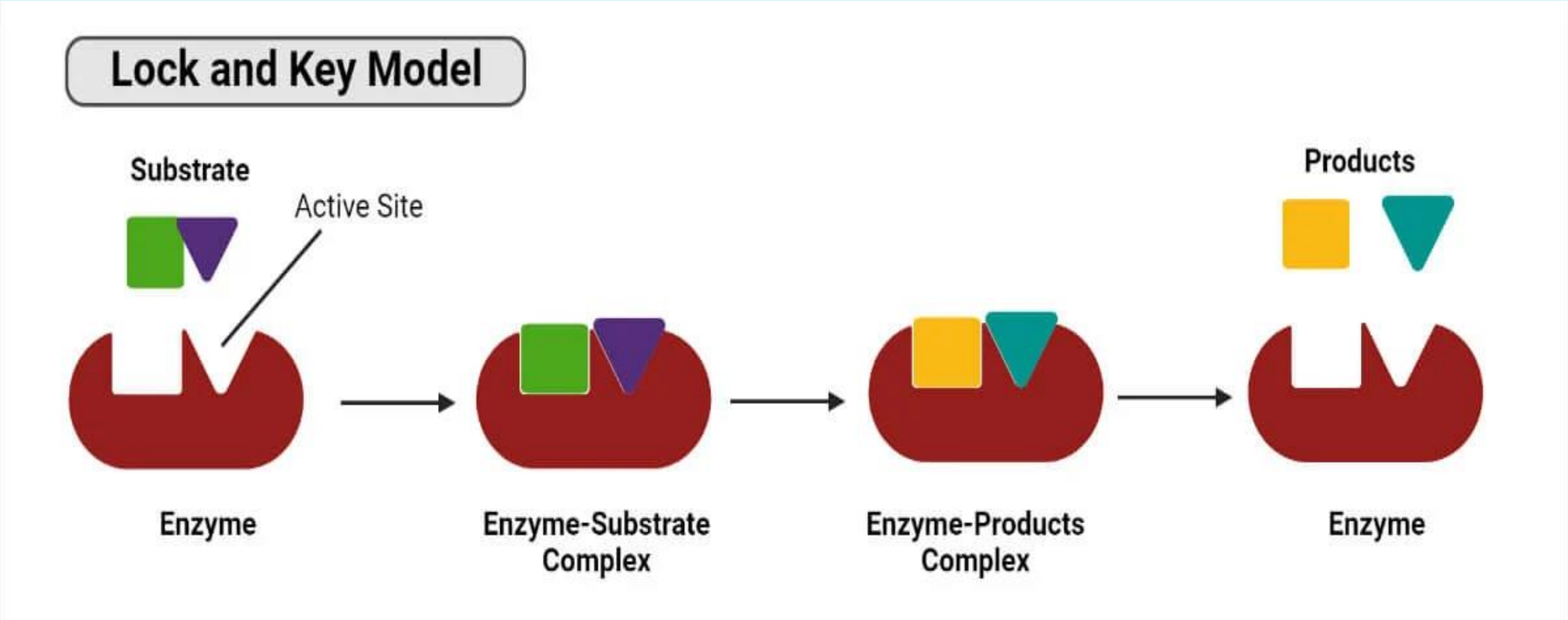
Models of Enzyme-Substrate Interaction

Lock-and-Key Model (Early Model)

- This model suggested that the enzyme's active site is a rigid structure perfectly complementary to the substrate, like a lock and key.

- While useful for illustrating specificity, it's an oversimplification.

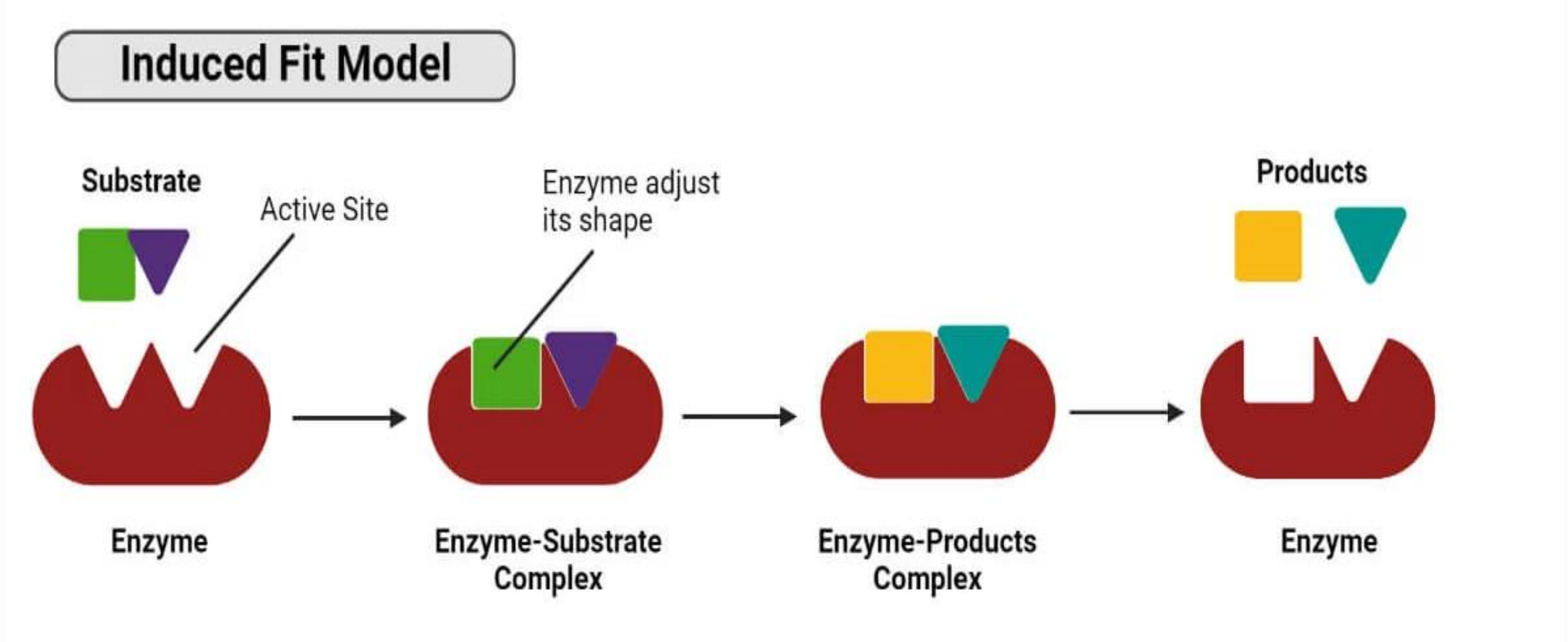

(a)
Key (substrate)
Lock (enzyme)
Substrate
Active site
Enzyme
(b)
Lock-Key Complex
Enzyme-Substrate
Complex


Lock and Key Model
Substrate
Active Site
Products
Enzyme
Enzyme-Substrate Complex
Enzyme-Products Complex
Enzyme

Models of Enzyme-Substrate Interaction

Induced-Fit Model (Current Model)

- This more accurate model states that both the enzyme and the substrate undergo slight conformational changes upon binding.
- The enzyme's active site molds itself around the substrate, inducing a tighter fit and optimizing the catalytic process.
- This dynamic interaction is crucial for catalysis.


Induced Fit Model
Substrate
Active Site
Enzyme adjust its shape
Products
Enzyme
Enzyme-Substrate Complex
Enzyme-Products Complex
Enzyme

Cofactors

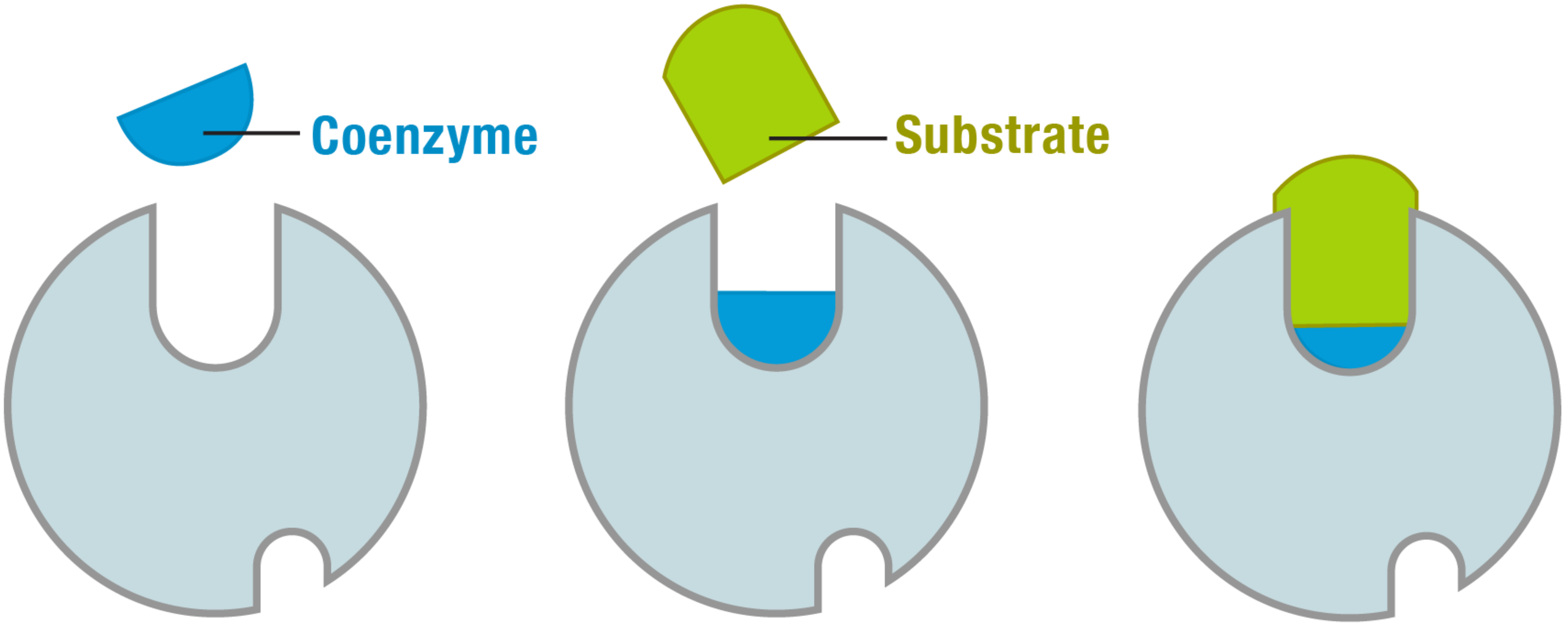
Many enzymes require additional non-protein components for their activity. These are broadly called **cofactors**.

Cofactors:

- **Inorganic ions:** Metal ions such as Mg2+, Zn2+, Fe2+, Cu2+ (e.g., DNA polymerase requires Mg2+).

- **Co-enzymes:** Complex organic molecules, often derived from vitamins. They act as transient carriers of specific functional groups (e.g., electrons, atoms, or small molecules).

  Examples: **NAD+** (nicotinamide adenine dinucleotide) and **FAD** (flavin adenine dinucleotide) are electron carriers; **Coenzyme A** (CoA) carries acyl groups; **ATP** carries phosphate groups.



Inactive enzyme

Coezyme activates enzyme

Active enzyme-coenzyme-substrate complex

Cofactors and Coenzymes

Terminology

- **Apoenzyme:** The inactive protein component of an enzyme, lacking its necessary cofactor.
- **Holoenzyme:** The catalytically active enzyme, consisting of the apoenzyme complexed with its cofactor.
- **Prosthetic Group:** A coenzyme (or cofactor) that is very tightly or covalently bound to the apoenzyme. (e.g., Heme in catalase).

Classification and Nomenclature of Enzymes

Systematic Nomenclature (EC Numbers)

- Each enzyme is assigned a unique **EC number**, consisting of four numbers separated by dots (e.g., EC 2.7.1.1).
- The first digit classifies the enzyme into one of six major classes based on the **type of reaction it catalyzes**.
- The subsequent digits further define the reaction, sub-class, and specific substrate.

Six Major Classes of Enzymes

- EC 1: Oxidoreductases
- EC 2: Transferases
- EC 3: Hydrolases
- EC 4: Lyases
- EC 5: Isomerases
- EC 6: Ligases

EC 1: Oxidoreductases

- **Function:** Catalyze oxidation-reduction reactions (transfer of electrons or hydrogen atoms).
- **Common Names:** Dehydrogenases, Oxidases, Reductases, Peroxidases.
- **Example: Lactate Dehydrogenase** (converts lactate to pyruvate, reducing NAD+ to NADH).

EC 2: Transferases

- **Function:** Catalyze the transfer of a functional group (e.g., methyl, phosphate, glycosyl) from one molecule to another.
- **Common Names:** Kinases (transfer phosphate), Transaminases (transfer amino groups).
- **Example: Hexokinase** (transfers a phosphate group from ATP to glucose).

EC 3: Hydrolases

- **Function:** Catalyze hydrolysis reactions – cleavage of bonds by the addition of water.
- **Common Names:** Proteases (cleave peptide bonds), Lipases (cleave ester bonds in lipids), Amylases (cleave glycosidic bonds in starch), Nucleases (cleave phosphodiester bonds in nucleic acids).
- **Example: Trypsin** (a protease that hydrolyzes peptide bonds in proteins).

EC 4: Lyases

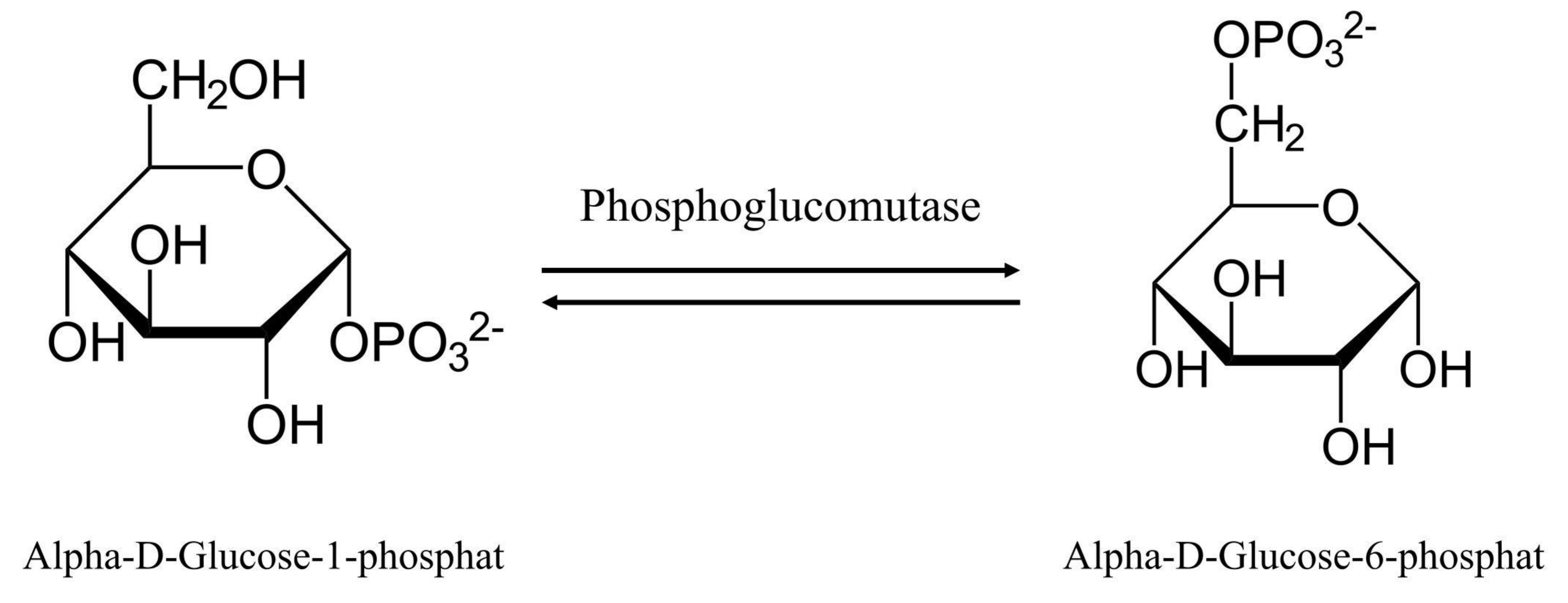
- **Function:** Catalyze the cleavage of C-C, C-O, C-N, or other bonds by means other than hydrolysis or oxidation, often forming double bonds or rings in the process. They add groups to or remove groups from double bonds.
- **Common Names:** Decarboxylases, Aldolase, Fumarase.
- **Example: Fumarase** (converts fumarate to malate by adding water across a double bond, but distinct from hydrolases in its mechanism).

EC 5: Isomerases

- **Function:** Catalyze intramolecular rearrangements – conversion of one isomer to another (e.g., structural isomers, geometric isomers, optical isomers).

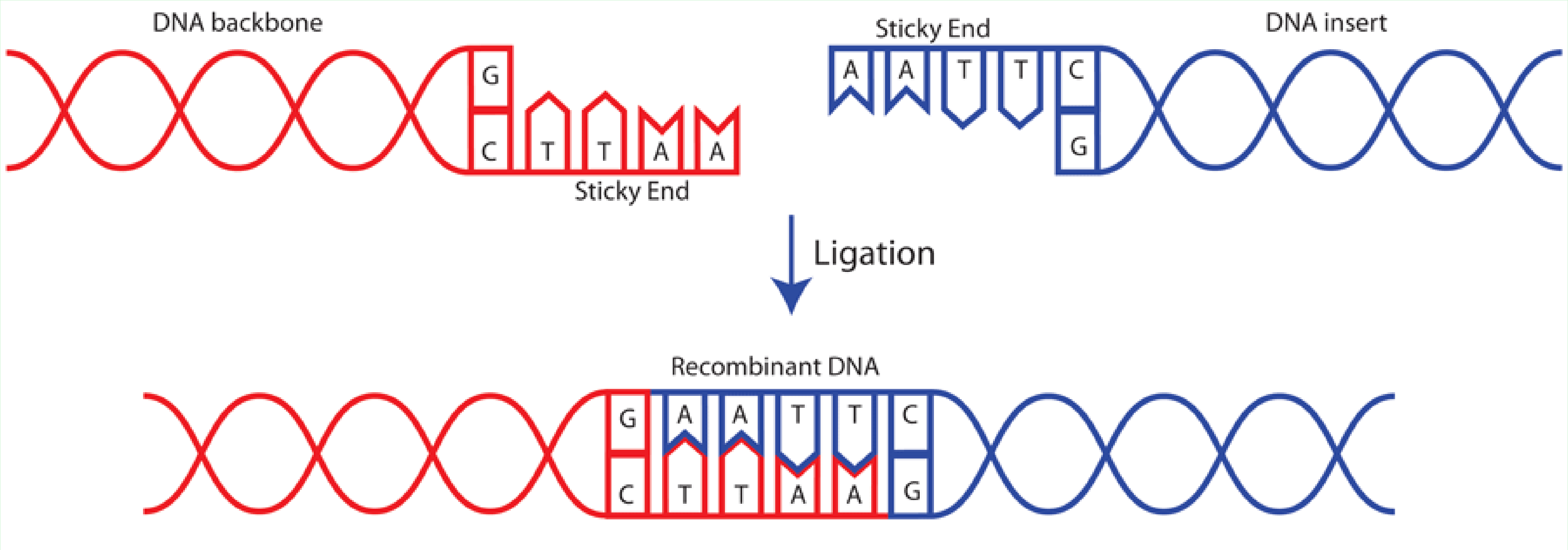
- **Common Names:** Mutases, Racemases, Epimerases.

- **Example:** **Phosphoglucomutase** (interconverts Glucose-1-phosphate and Glucose-6-phosphate).


CH2OH
OH
OH
OPO3 2-
OH
Phosphoglucomutase
OPO3 2-
CH2
OH
OH
OH
OH
Alpha-D-Glucose-1-phosphat
Alpha-D-Glucose-6-phosphat

EC 6: Ligases

- **Function:** Catalyze the formation of new bonds (e.g., C-C, C-S, C-O, C-N) by coupling the reaction to the hydrolysis of ATP or another energy-rich compound. They "ligate" or join two molecules together.
- **Common Names:** Synthetases (implies ATP hydrolysis), Carboxylases.
- **Example: DNA Ligase** (joins DNA strands, requiring ATP).


DNA backbone
G
C
T
T
A
A
Sticky End
Sticky End
DNA insert
A
A
T
T
C
G
Ligation
Recombinant DNA
G
A
A
T
T
C
C
T
T
A
A
G

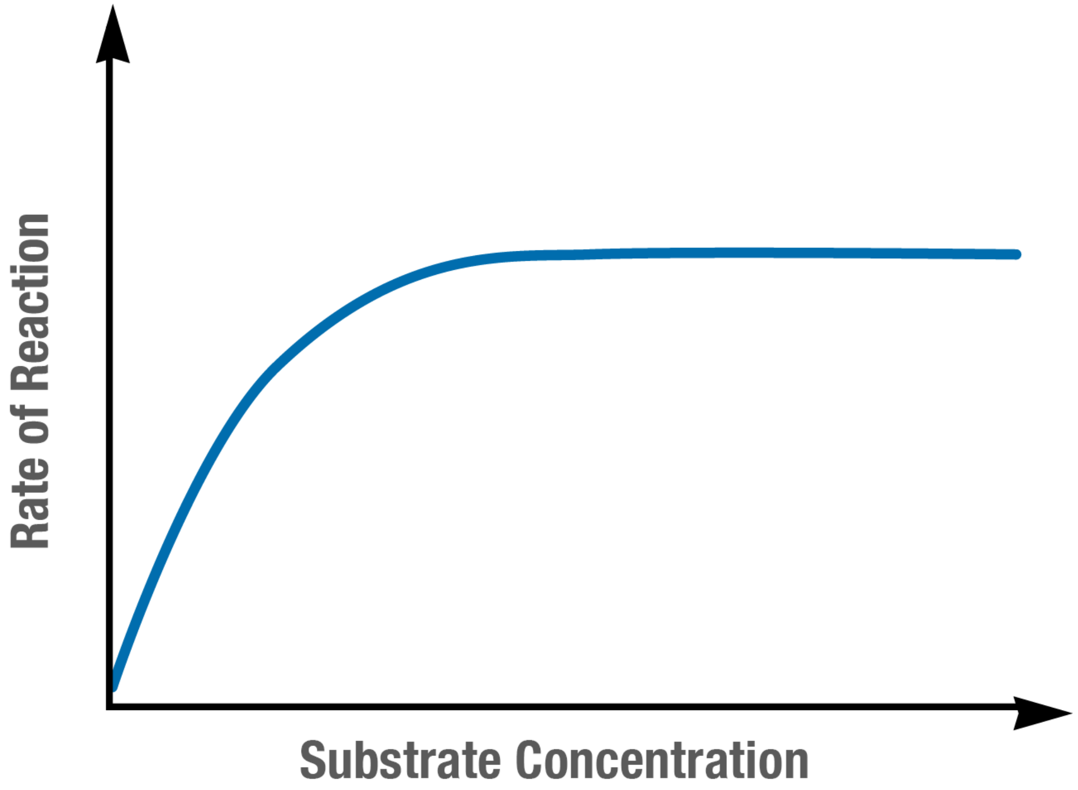
Kinetic Analysis of Enzyme-Catalyzed Reactions

Factors Affecting Enzyme Activity:

- Substrate Concentration ([S])
- Enzyme Concentration ([E])
- Temperature
- pH
- Inhibitors and Activators

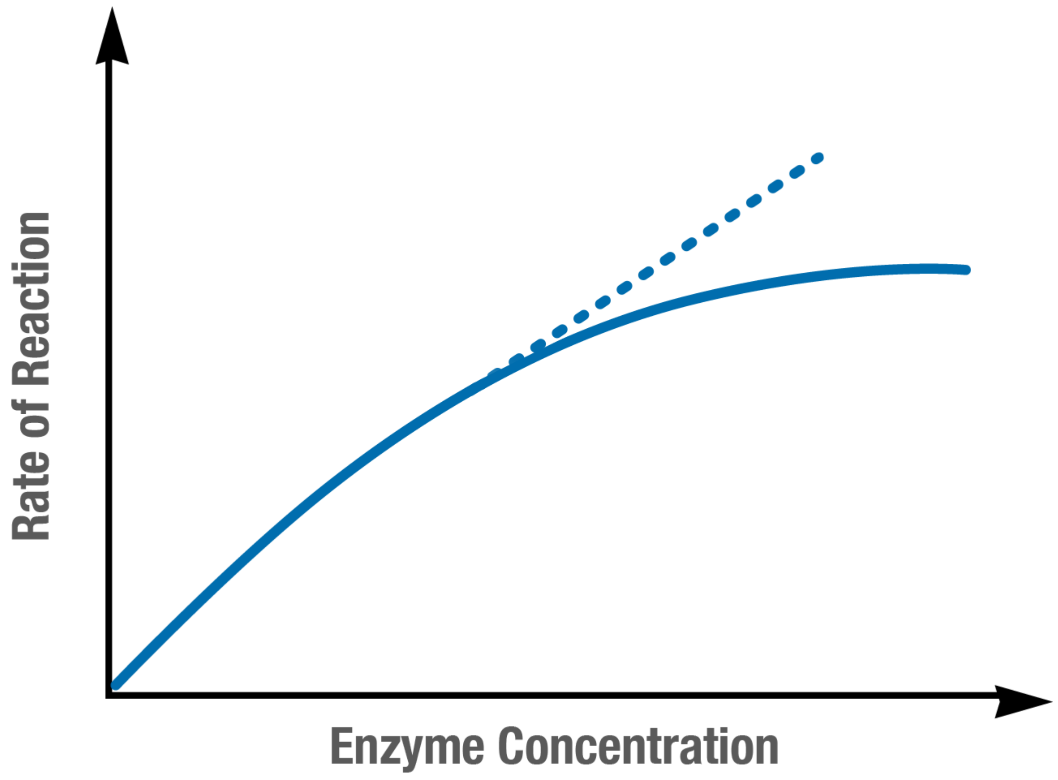
Substrate Concentration ([S])

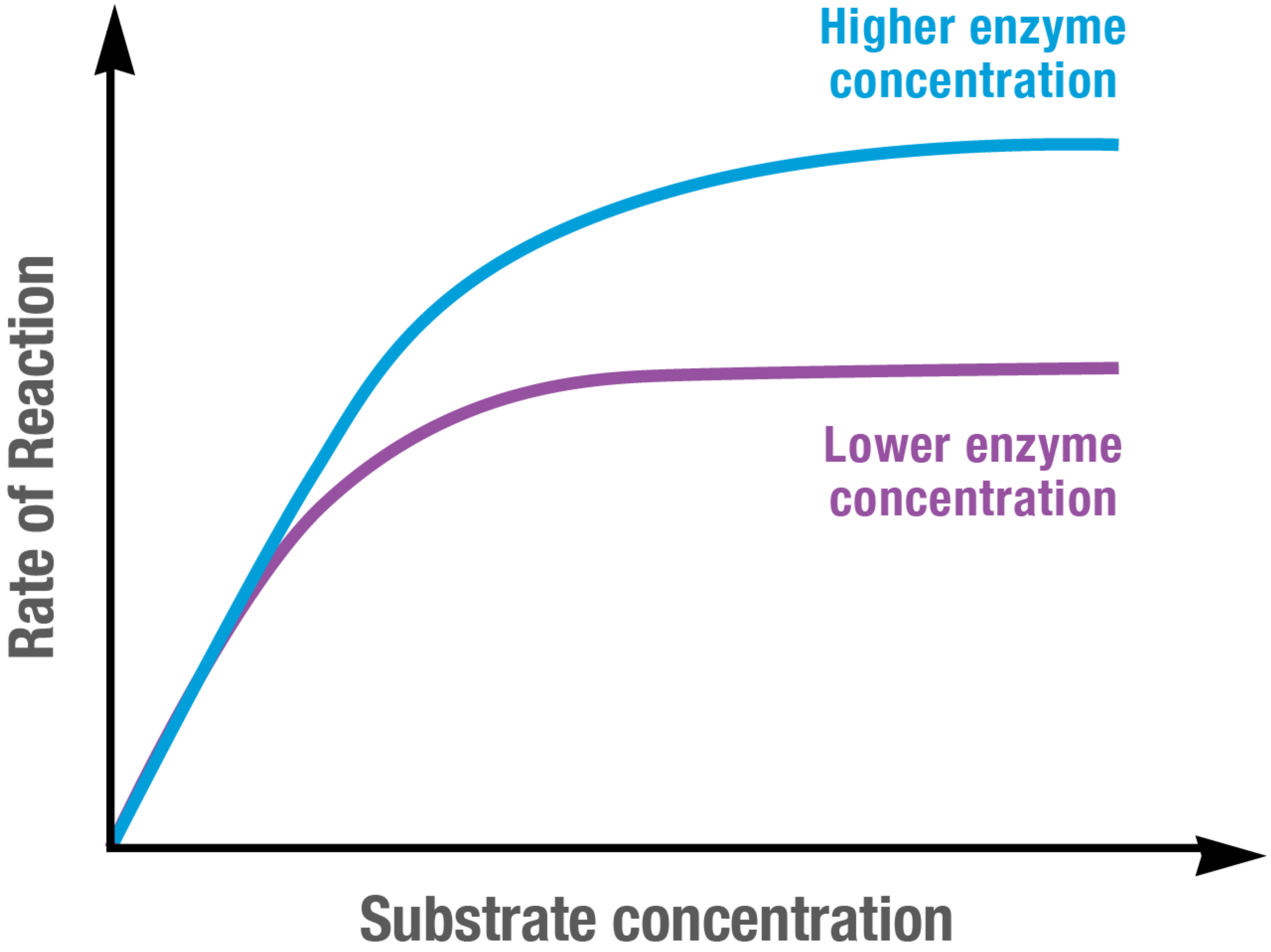
- As [S] increases, the reaction rate generally increases until the enzyme becomes saturated.



Enzyme Concentration ([E])

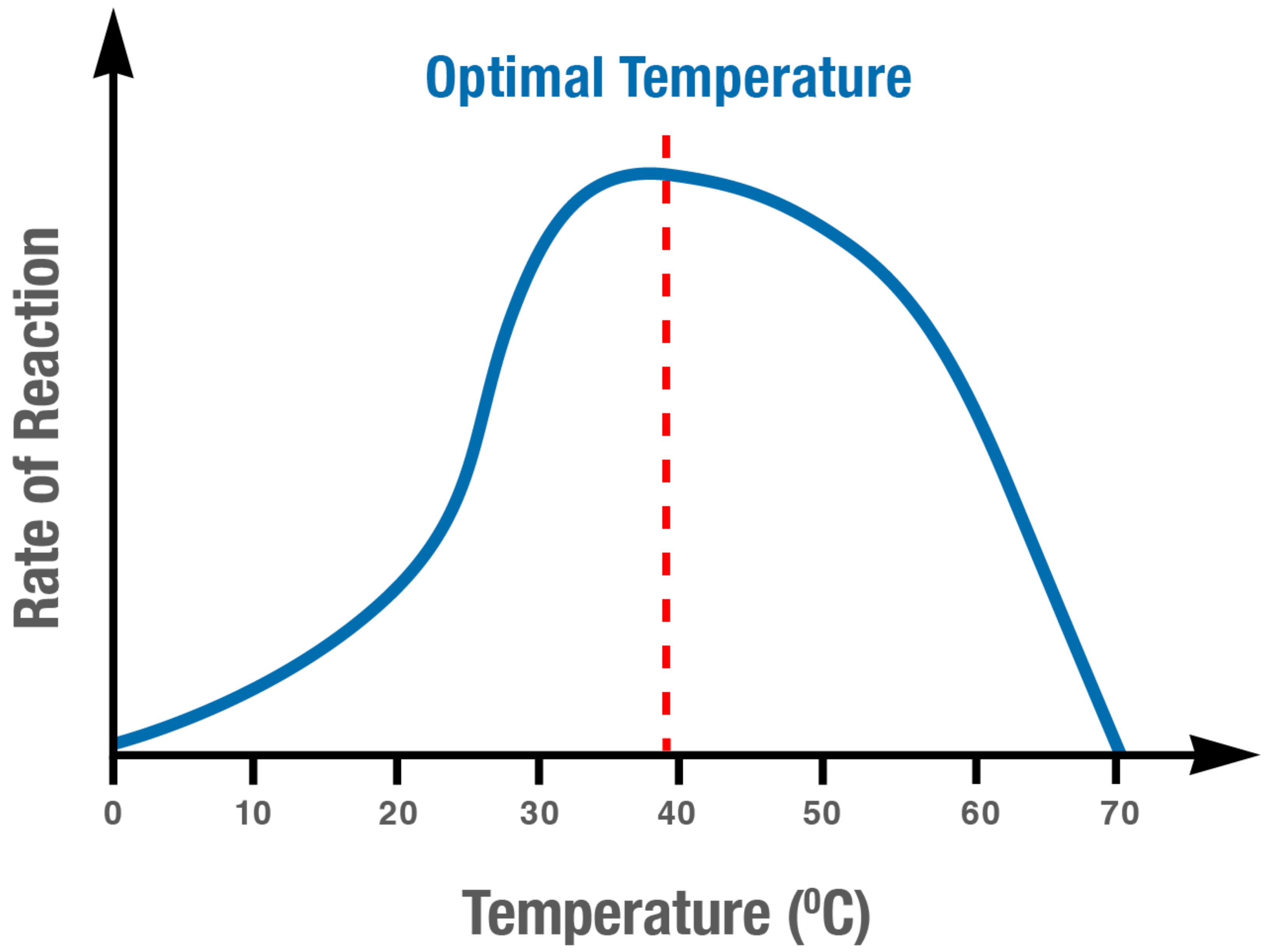
- The reaction rate is directly proportional to [E], assuming [S] is saturating.




Higher enzyme concentration
Lower enzyme concentration
Rate of Reaction
Substrate concentration

Temperature

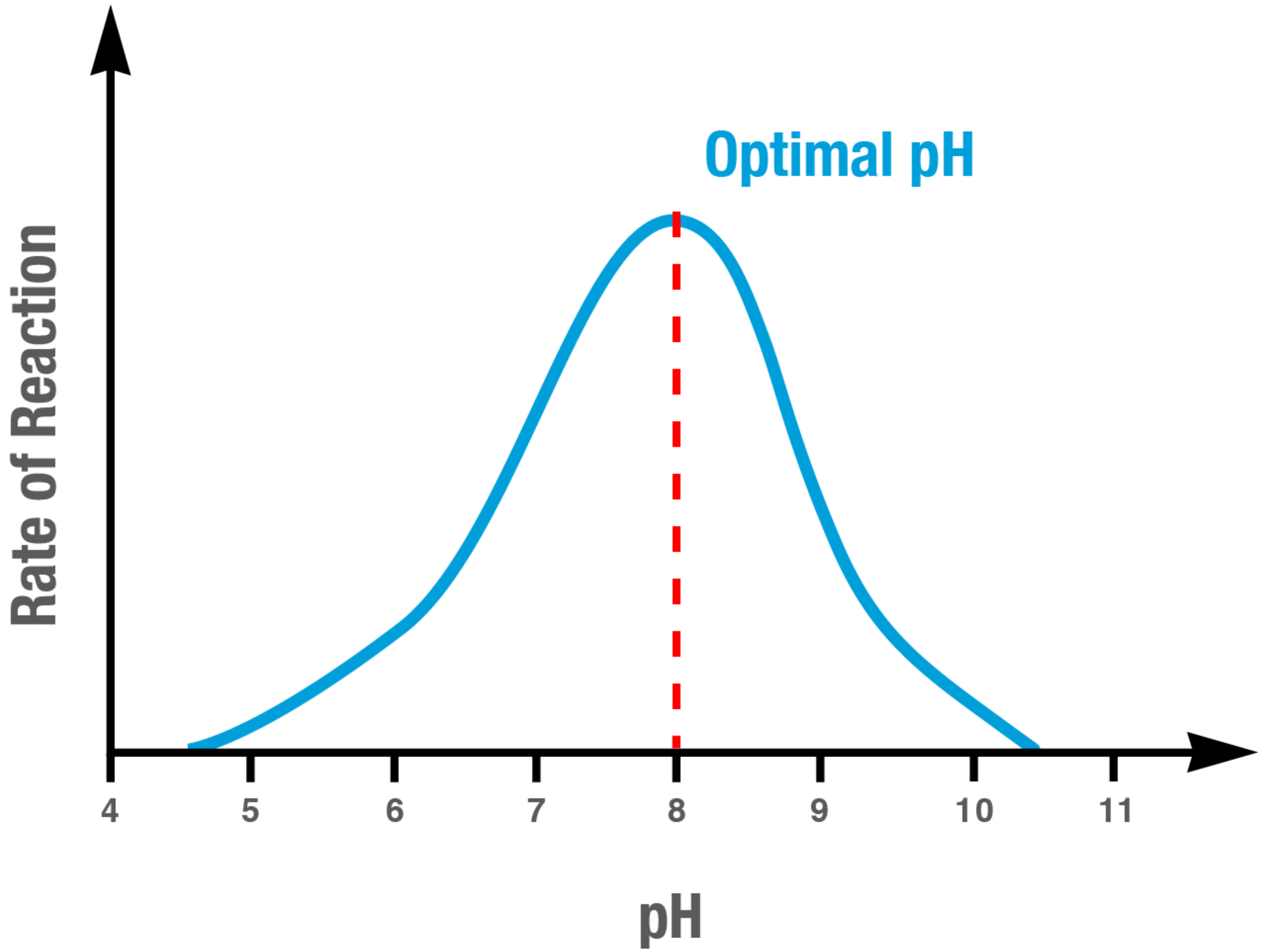
- Increasing temperature generally increases reaction rate (molecules move faster, more collisions).
- However, beyond an optimal temperature, enzymes rapidly lose activity due to **denaturation** (loss of 3D structure).

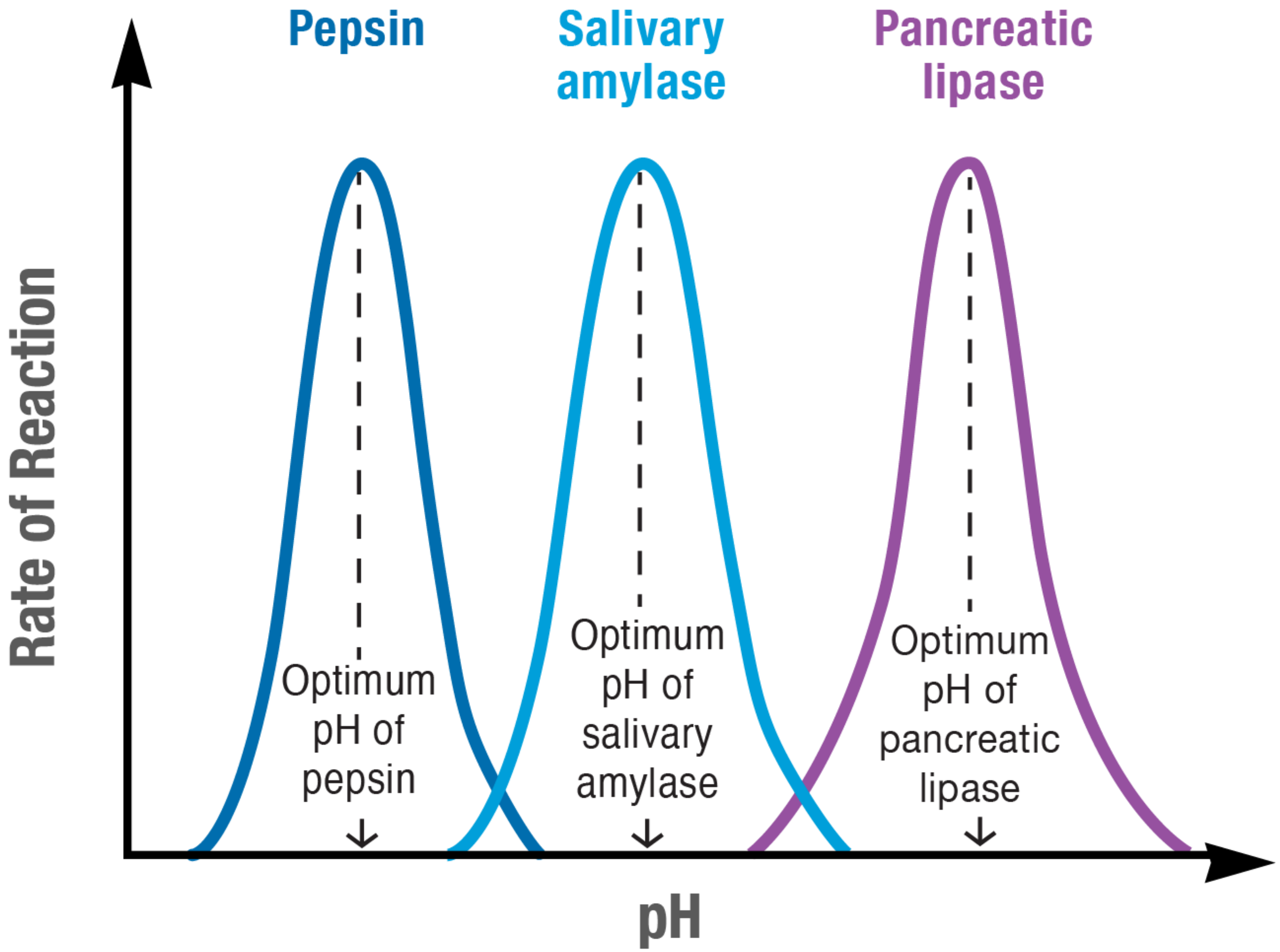


Optimal Temperature
Rate of Reaction
0
10
20
30
40
50
60
70
Temperature (^{0}C)

pH

- Each enzyme has an optimal pH at which its activity is maximal.
- Extreme pH values can alter the ionization state of active site amino acids and cause denaturation, leading to loss of activity.
- (e.g., Pepsin (stomach) optimal pH ~2, Trypsin (small intestine) optimal pH ~8).

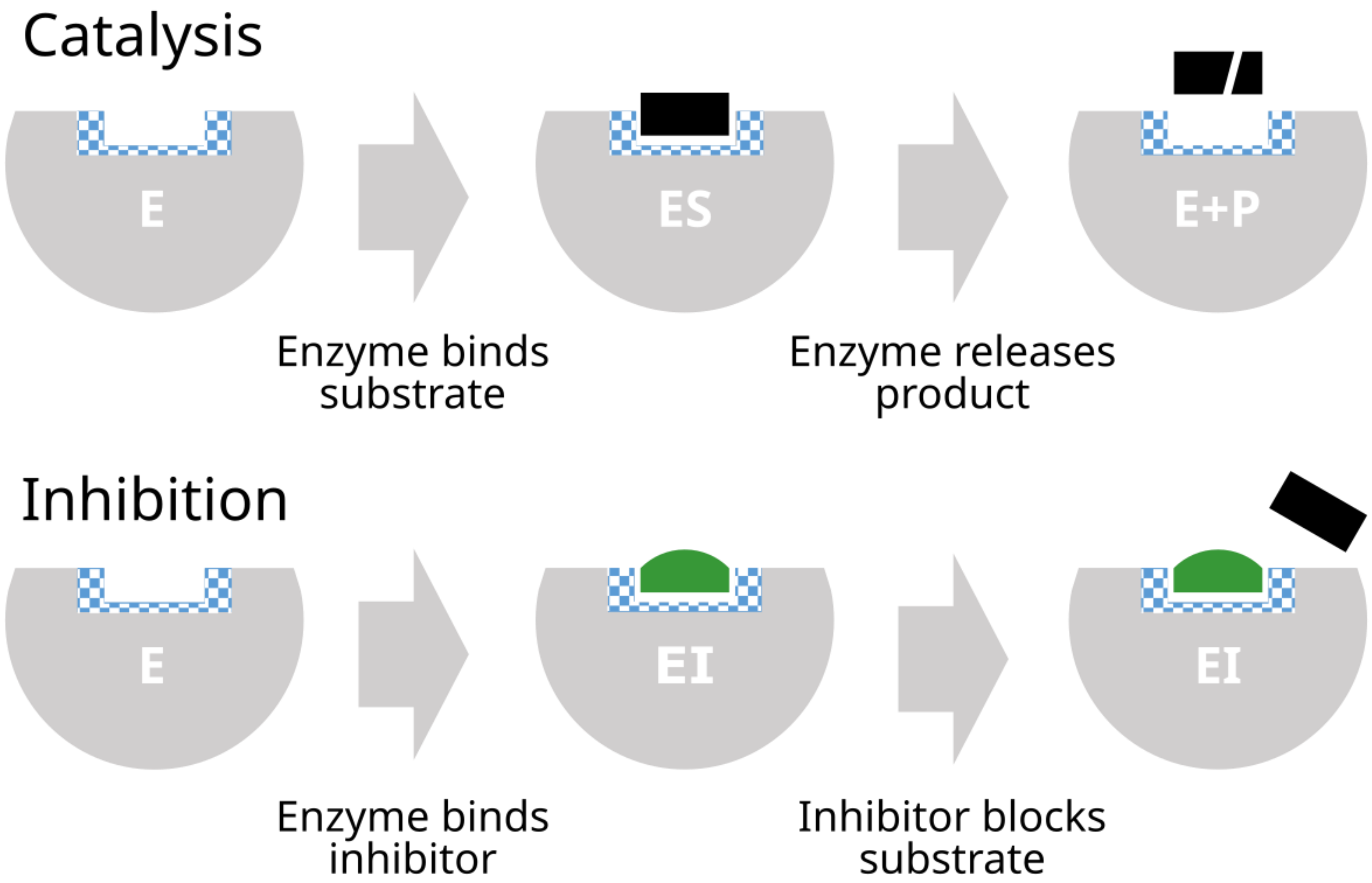

Optimal pH
Rate of Reaction
4
5
6
7
8
9
10
11
pH



Pepsin
Salivary amylase
Pancreatic lipase
Rate of Reaction
Optimum pH of pepsin
Optimum pH of salivary amylase
Optimum pH of pancreatic lipase
pH

Inhibitors and Activators

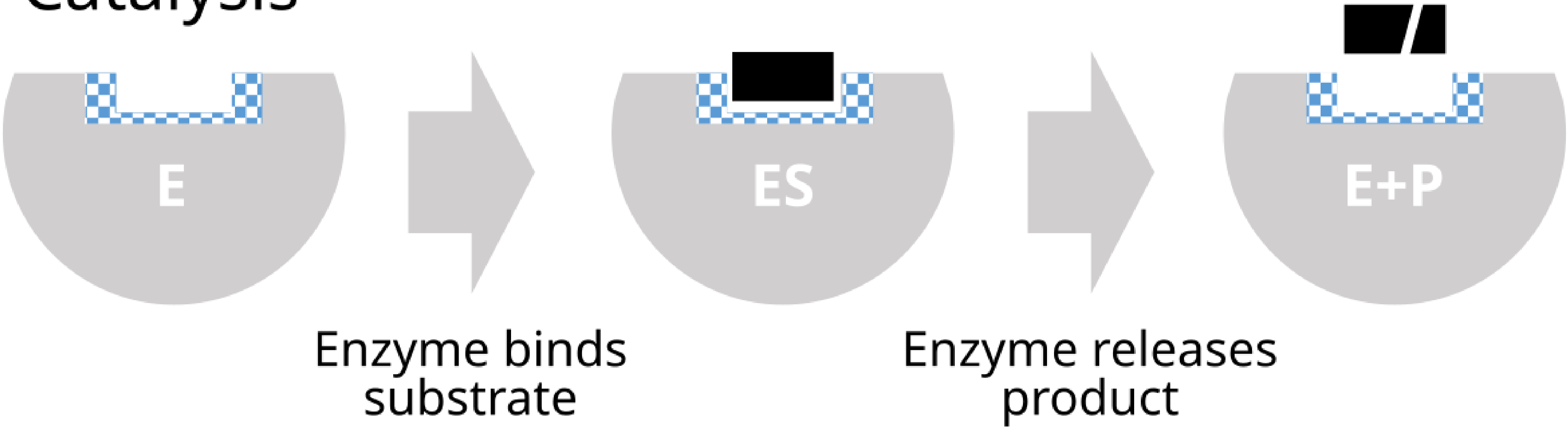
- Molecules that decrease or increase enzyme activity



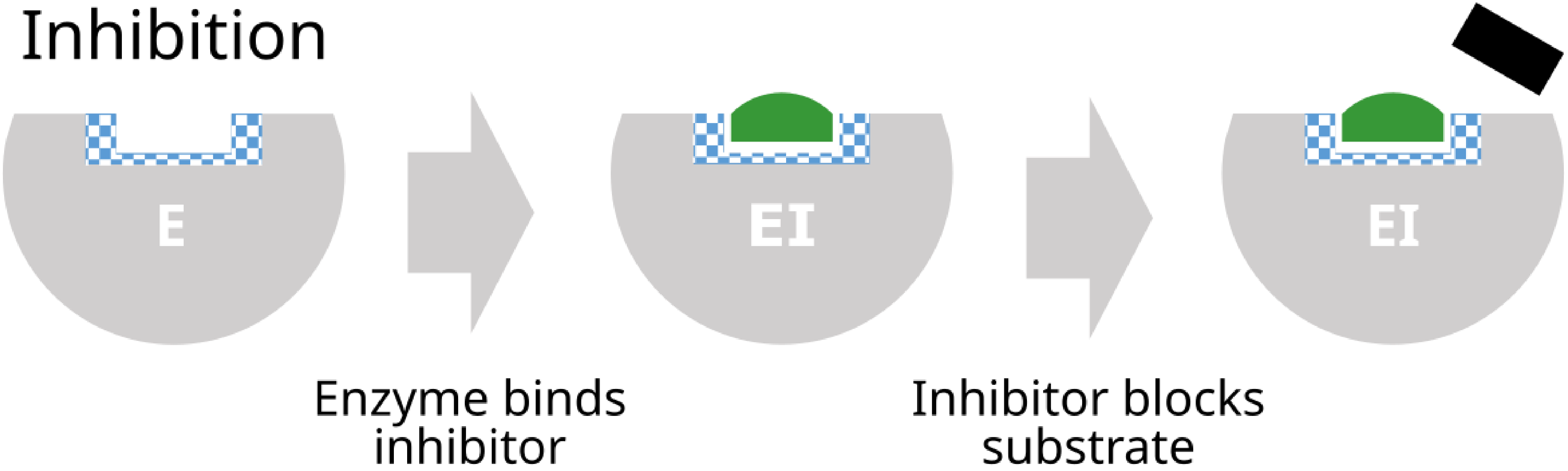
Enzyme Inhibition

- **Enzyme inhibitors** are molecules that decrease the activity of an enzyme.
- They are crucial in metabolic regulation and are important targets for drug design.

Catalysis



Inhibition



Reversible Inhibition

The inhibitor binds non-covalently to the enzyme and can dissociate, allowing the enzyme to regain activity.

Competitive Inhibition:

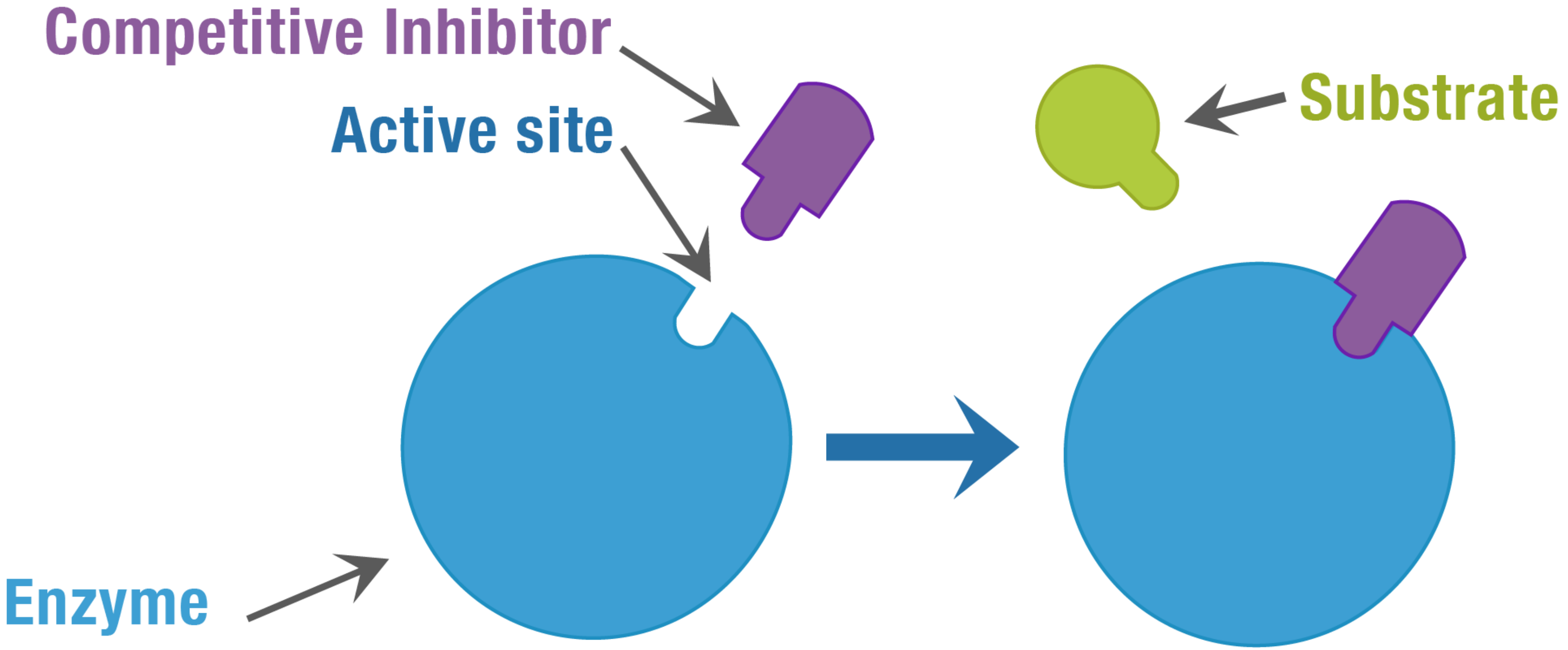
- **Mechanism:** The inhibitor resembles the substrate and competes with the substrate for binding to the **active site**.

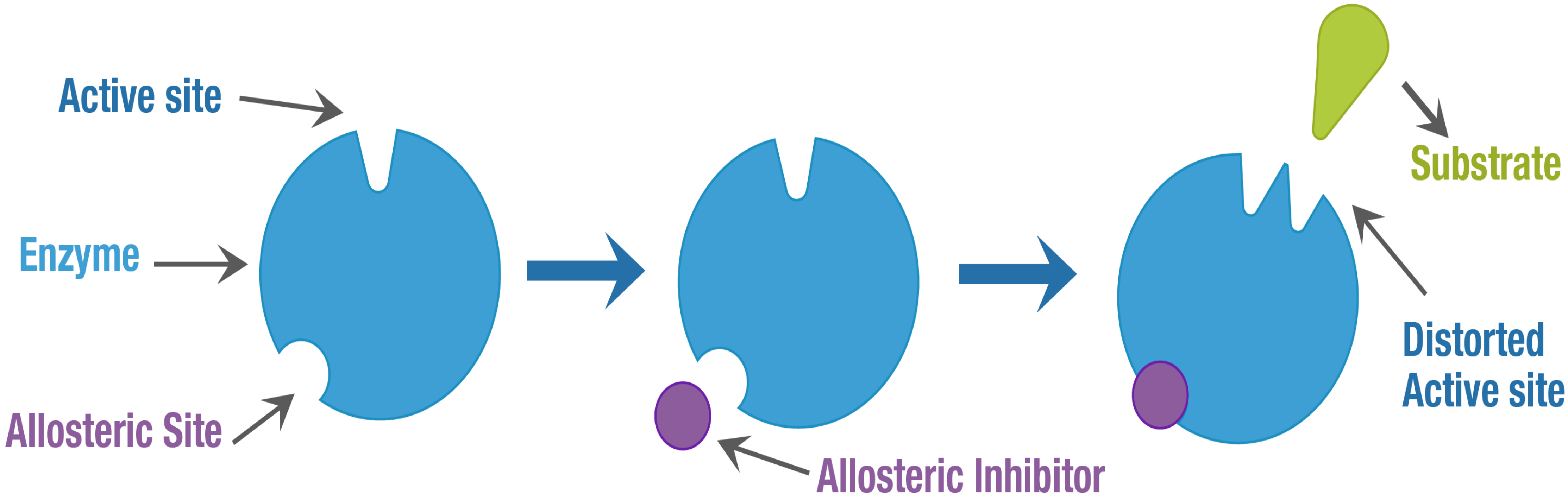
Uncompetitive Inhibition:

- **Mechanism:** The inhibitor binds *only* to the **enzyme-substrate (ES) complex**, not to the free enzyme. It binds at a site distinct from the active site.

Non-competitive (Mixed) Inhibition:

- **Mechanism:** The inhibitor binds to a site distinct from the active site, and can bind to either the **free enzyme (E) or the ES complex**. It causes a conformational change that reduces catalytic efficiency.


Competitive Inhibitor
Active site
Substrate
Enzyme


Active site
Enzyme
Allosteric Site
Allosteric Inhibitor
Substrate
Distorted
Active site

Irreversible Inhibition

The inhibitor forms a stable, often covalent, bond with the enzyme, permanently inactivating it.

Mechanism: Often involves modification of an amino acid residue in the active site or a site crucial for catalysis.

Examples:

- **Heavy metals:** (e.g., lead, mercury) can bind to sulfhydryl groups of cysteine residues.
- **Nerve gases:** (e.g., sarin) inhibit acetylcholinesterase, a crucial enzyme in nerve impulse transmission.
- **Aspirin:** Irreversibly inhibits cyclooxygenase (COX) enzymes, reducing inflammation.

Regulation of Enzyme Activity by Non-Genetic Mechanisms

- Cells need to control enzyme activity dynamically to respond to changing metabolic needs.

- This regulation occurs at various levels, beyond just controlling enzyme synthesis (genetic regulation).

Allosteric Regulation

Definition: The regulation of an enzyme's activity by the binding of an effector molecule at a site **other than the active site** (this non-active site is called the **allosteric site** or regulatory site).

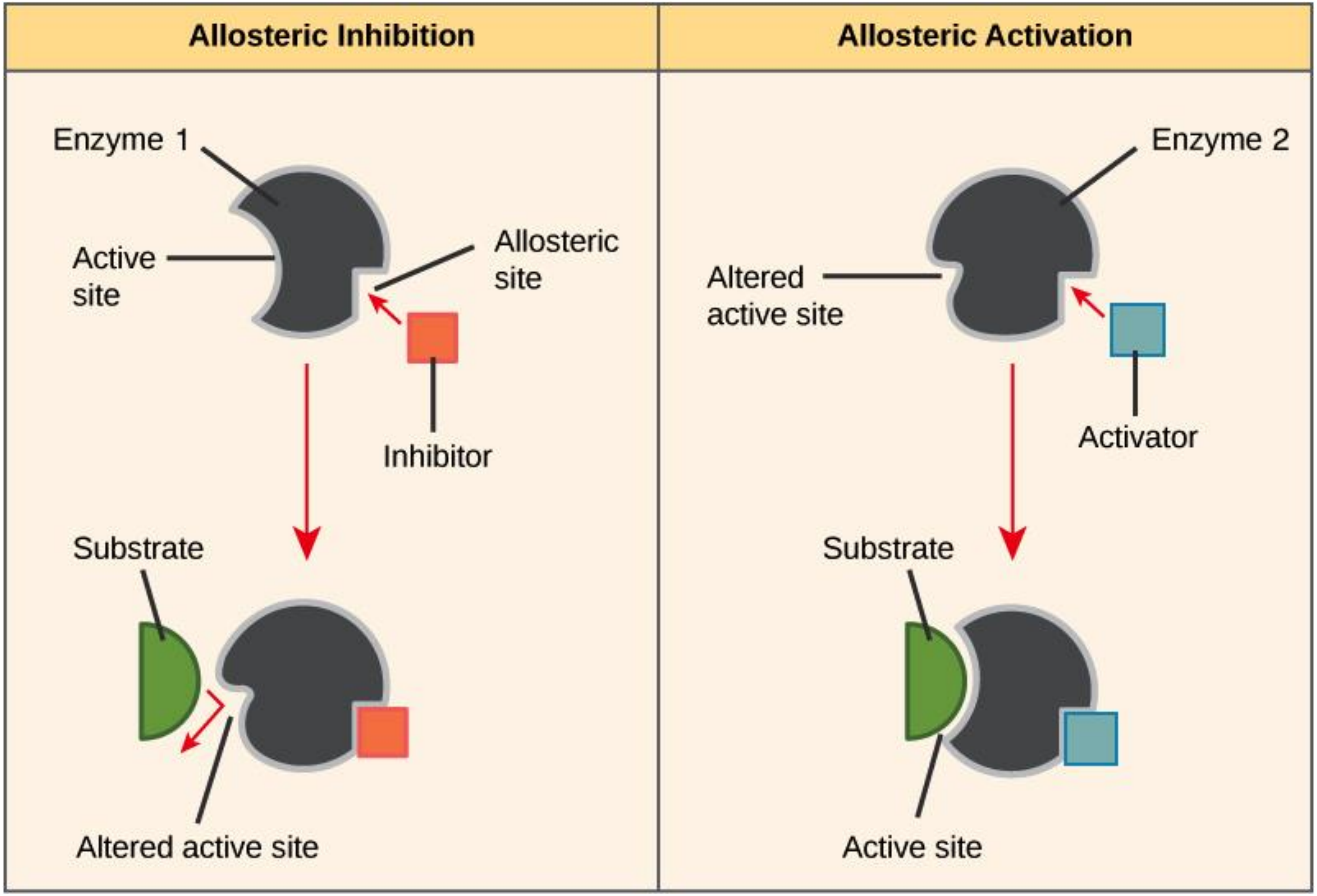
Allosteric Enzymes

- Are typically **multi-subunit proteins**.
- Often display **cooperative binding** (binding of a substrate or effector to one subunit affects the binding affinity or catalytic activity of other subunits).

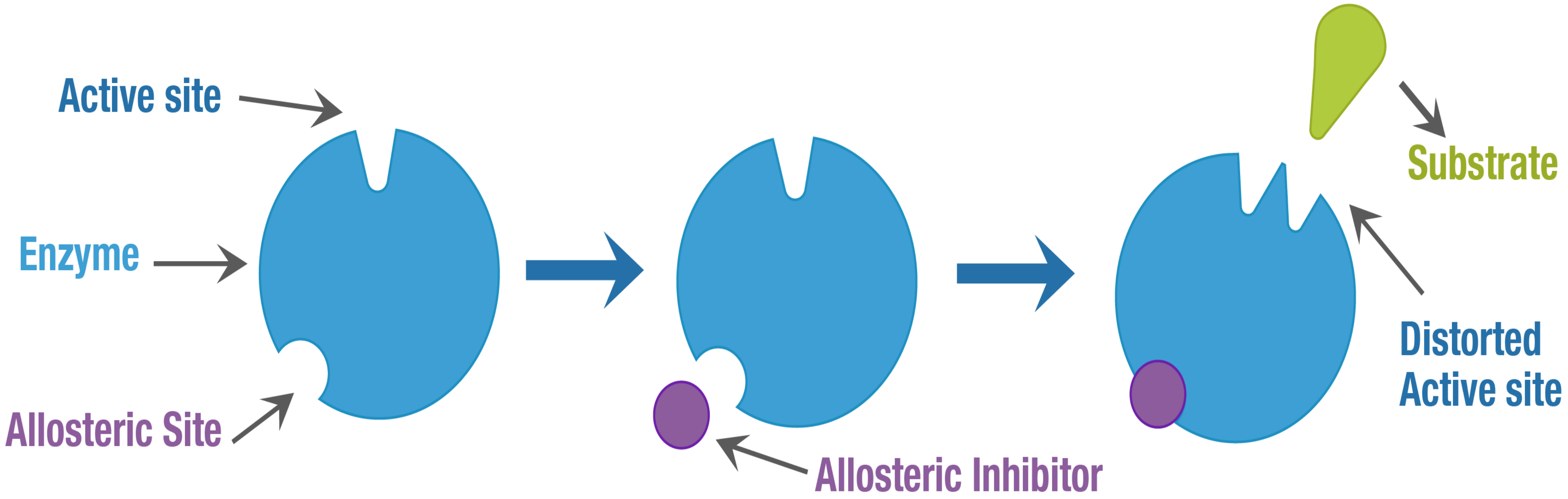
Allosteric Effectors (Modulators)

- **Allosteric Activators:** Bind to an allosteric site and shift the enzyme into a more active conformation.

- **Allosteric Inhibitors:** Bind to an allosteric site and shift the enzyme into a less active conformation.



Allosteric Inhibition
Allosteric Activation
Enzyme 1
Active site
Allosteric site
Inhibitor
Substrate
Altered active site
Enzyme 2
Altered active site
Activator
Substrate
Active site



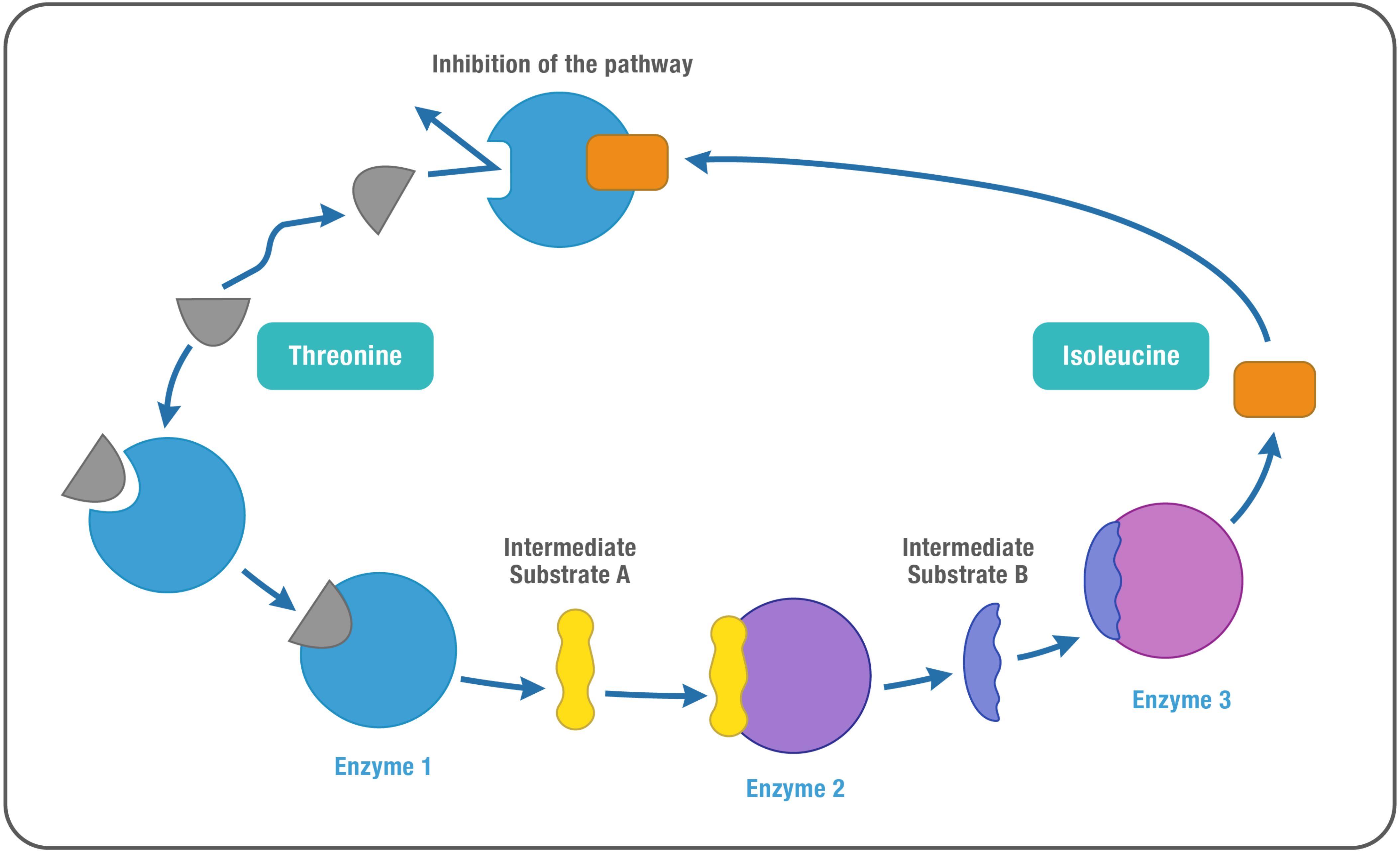
Active site
Enzyme
Allosteric Site
Allosteric Inhibitor
Substrate
Distorted
Active site

Feedback Inhibition (End-product Inhibition)

- A common allosteric regulatory mechanism where the end product of a metabolic pathway inhibits an enzyme (often the first committed step) earlier in the same pathway.

- This prevents wasteful overproduction of the end product.

  Example: In the synthesis of ATP, high levels of ATP (the end product) inhibit enzymes in glycolysis (e.g., PFK-1) and the citric acid cycle, slowing down ATP production when energy is abundant.


Inhibition of the pathway
Threonine
Isoleucine
Intermediate Substrate A
Intermediate Substrate B
Enzyme 1
Enzyme 2
Enzyme 3

Regulation by Reversible Covalent Modification

- **Definition:** Enzyme activity is regulated by the reversible **covalent attachment or removal of a small chemical group** to specific amino acid residues on the enzyme protein.

- This can cause a conformational change that alters activity.

Phosphorylation/ Dephosphorylation

- A **phosphate group** (from ATP) is covalently attached to the hydroxyl groups of **serine (Ser), threonine (Thr), or tyrosine (Tyr)** residues on the enzyme.

- **Protein Kinases:** Enzymes that catalyze the addition of phosphate groups (phosphorylation).
- **Protein Phosphatases:** Enzymes that catalyze the removal of phosphate groups (dephosphorylation).

- **Effect:** Phosphorylation can either **activate** or **inactivate** an enzyme, depending on the specific enzyme. This provides a rapid and reversible "on/off" switch for many cellular processes.

  - **Example: Glycogen Phosphorylase** (breaks down glycogen) is *activated* by phosphorylation. **Glycogen Synthase** (synthesizes glycogen) is *inactivated* by phosphorylation. This allows for reciprocal regulation of opposing pathways.

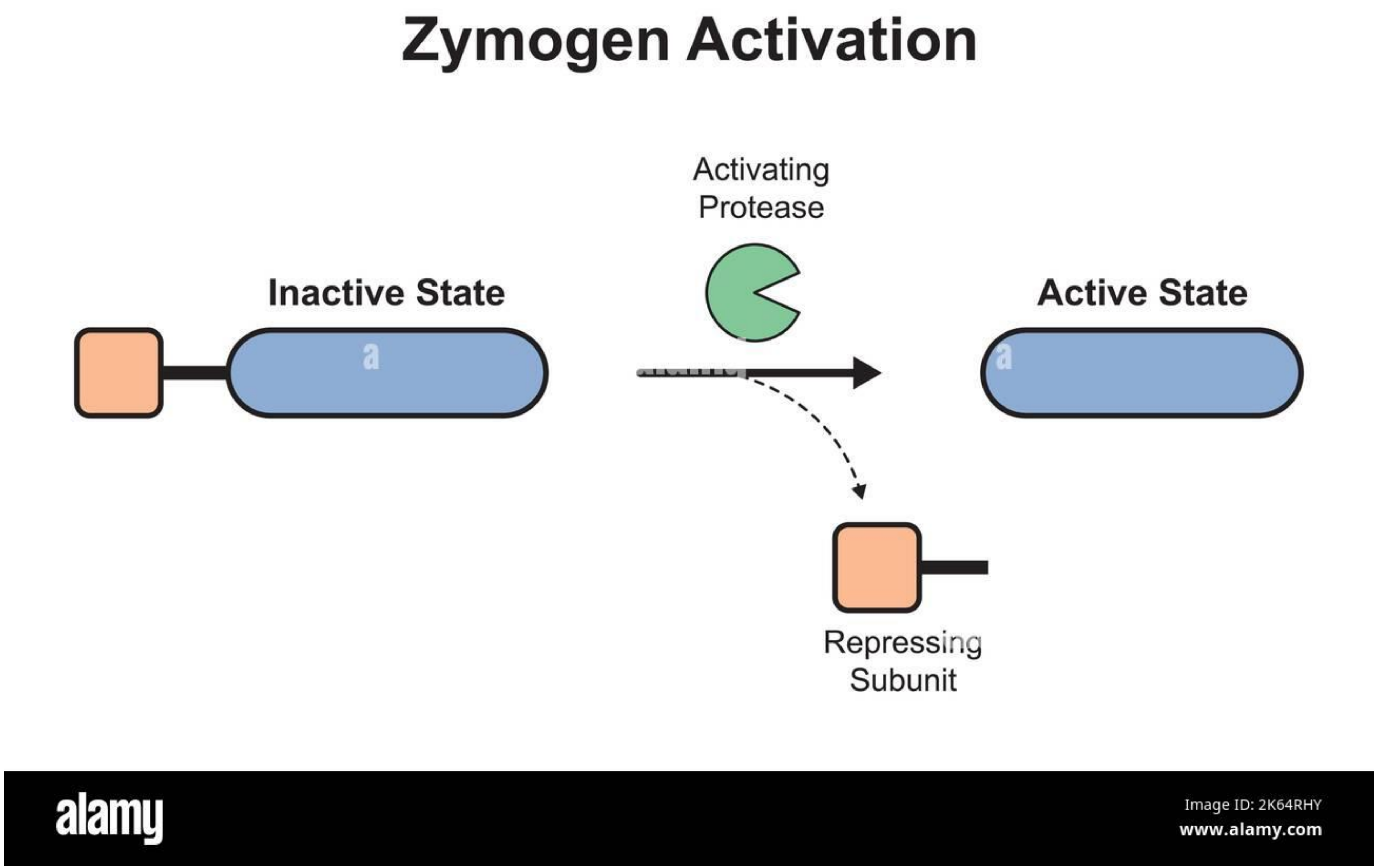
- **Other Covalent Modifications:** Acetylation (e.g., histones), methylation, glycosylation, ubiquitination (targeting for degradation), ADP-ribosylation.

Regulation by Proteolytic Activation (Zymogens / Proenzymes)

Definition: Many enzymes are synthesized as inactive precursors called **zymogens** (or **proenzymes**).

They are activated by **irreversible proteolytic cleavage** of one or more peptide bonds, typically by another protease.

Purpose: This mechanism provides a way to control enzyme activity precisely in space and time, preventing premature or uncontrolled activity that could be damaging to the cell or organism.


Zymogen Activation
Activating
Protease
Inactive State
Active State
Repressing
Subunit
alamy
Image ID: 2K64RHY
www.alamy.com

Zymogens / Proenzymes

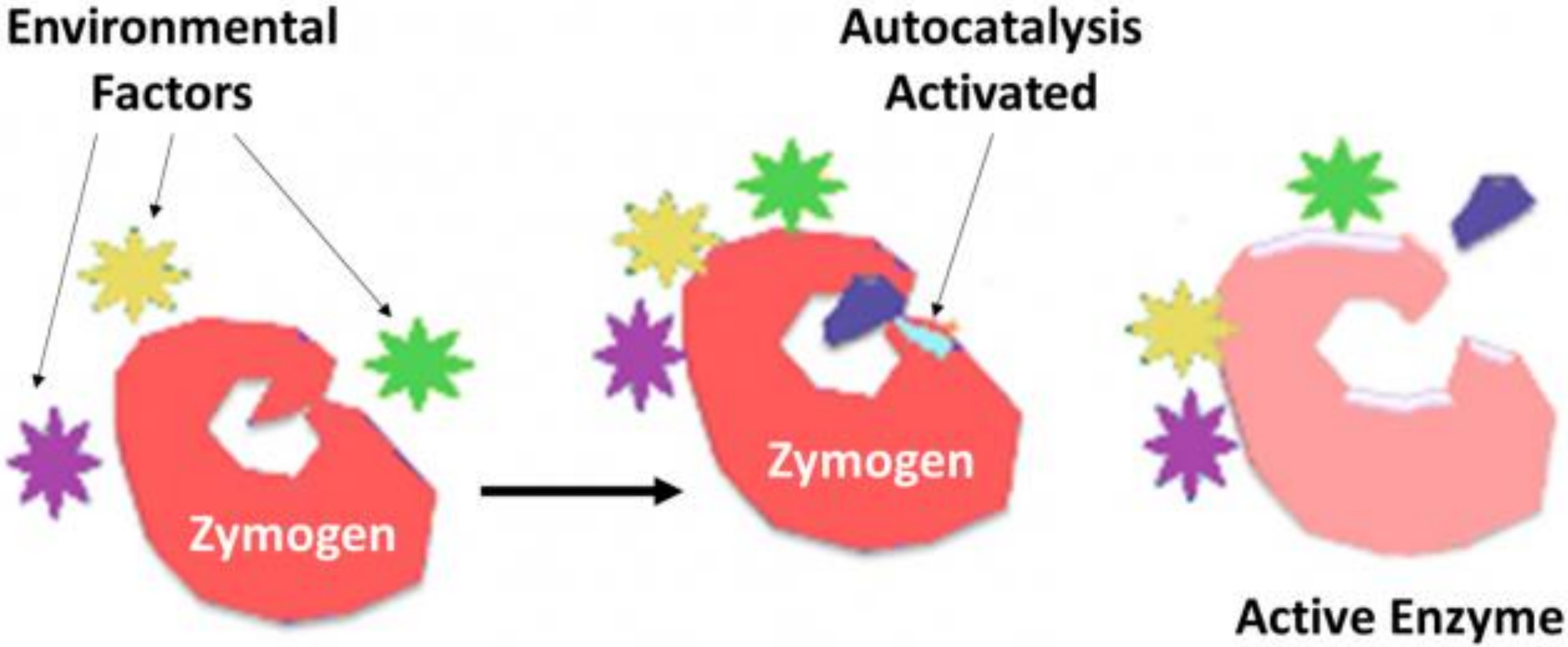
Examples:

- **Digestive Enzymes:** Synthesized as zymogens in the pancreas and stomach to prevent self-digestion.
  - **Pepsinogen** (inactive) is activated to **Pepsin** (active) by stomach acid.
  - **Trypsinogen** (inactive) is activated to **Trypsin** (active) by enteropeptidase in the small intestine. Trypsin then activates other digestive zymogens (e.g., chymotrypsinogen to chymotrypsin).

- **Blood Clotting Factors:** The clotting cascade involves a series of zymogen activations, ensuring that clotting only occurs rapidly and locally when needed.

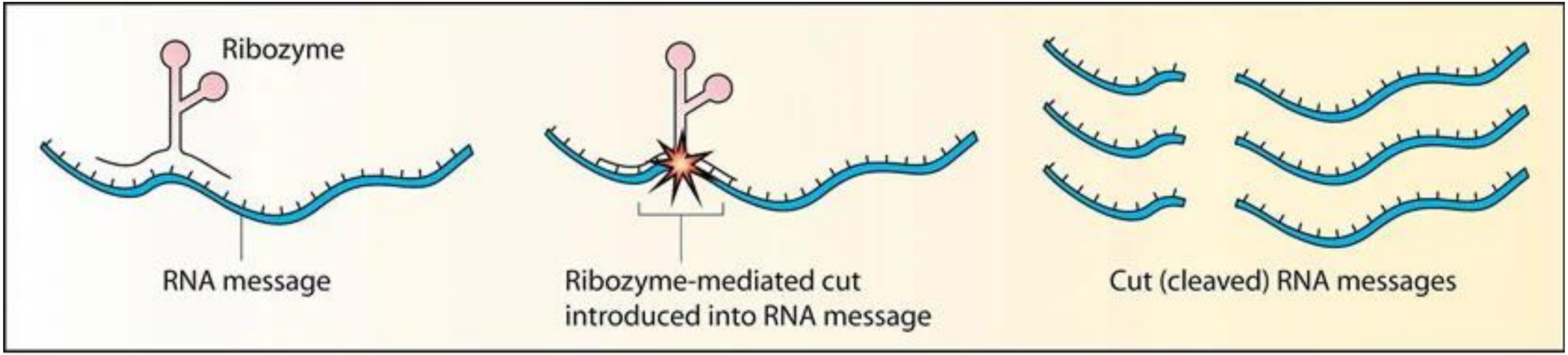
- **Hormones: Proinsulin** (inactive precursor) is cleaved to form active **Insulin**.

- **Collagenases:** Enzymes that break down collagen, synthesized as zymogens to prevent uncontrolled tissue degradation.


Environmental Factors
Zymogen
Autocatalysis Activated
Zymogen
Active Enzyme

Ribozymes: When RNA Acts as a Catalyst

- For a long time, it was believed that all enzymes were proteins.
- However, in the 1980s, the discovery of **ribozymes** challenged this dogma.
- **Definition: Ribozymes** are RNA molecules that possess catalytic activity. They can accelerate specific biochemical reactions, similar to protein enzymes.



Ribozyme Mediated Cleavage of RNA

Ribozymes

Examples:

- **Ribonuclease P:** An RNA-protein complex where the RNA component (M1 RNA) is the catalytic subunit responsible for cleaving precursors to transfer RNA (tRNA).
- **Ribosomal RNA (rRNA):** The large ribosomal RNA (rRNA) molecule within the ribosome is responsible for the **peptidyl transferase activity** that catalyzes the formation of peptide bonds during protein synthesis. This is a crucial catalytic function.
- **Self-splicing introns:** Certain RNA molecules (introns) can catalyze their own removal from larger RNA transcripts without the help of protein enzymes.

Significance: The discovery of ribozymes provided strong evidence for the "RNA world" hypothesis, suggesting that early life forms might have used RNA for both genetic information storage and catalysis, before the evolution of proteins.

Regulatory Enzymes: General Properties and Mechanisms

- Regulatory enzymes are those that control the flow of metabolic pathways.
- They are typically found at key control points, often at the first committed step of a pathway.

General Properties

- Often catalyze **irreversible** steps in a pathway.
- Tend to be **allosteric enzymes**, meaning their activity is modulated by effectors binding at sites other than the active site. This allows for rapid and sensitive responses to cellular signals.
- Are frequently targets for **covalent modification**, especially phosphorylation, providing a rapid and reversible "on/off" switch controlled by signaling cascades.
- Can be subject to **proteolytic activation (zymogen activation)** for precise, irreversible control in specific physiological contexts.
- Often **multi-subunit proteins** exhibiting cooperativity, which enhances their sensitivity to regulatory signals.

Interplay of Mechanisms

- It's common for a single regulatory enzyme to be controlled by multiple mechanisms.

- For example, **Phosphofructokinase-1 (PFK-1)** in glycolysis is a classic allosteric enzyme regulated by ATP, AMP, and citrate (allosteric effectors), and its activity is also influenced by phosphorylation state.

THANK YOU